

Debug the Cause

Project: Helpful Classroom Sorter

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Repeating Efficiently

Without opening the old lesson, explain one key idea from Repeating Efficiently. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Algorithms as Recipes

Without opening the old lesson, explain one key idea from Algorithms as Recipes. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Sound Becomes a Signal

Without opening the old lesson, explain one key idea from Sound Becomes a Signal. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Choose between two explanations for Debugging with revealing tests. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Debugging with revealing tests. Proposed explanation: “The last wrong number locates the fault”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Debug a new ticket rule

A paper ticket rule must add 1 to the input count. If the updated count is at least 5, report Group; otherwise report Single. A copied version says above 5. Choose the smallest three consecutive nonnegative starting counts that test just below, exactly at and just above the comparison boundary. Predict both versions, identify the revealing start and state the repair and retests.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. Your learning target

Starting at square 0, the target is square 3. Program: repeat twice {MOVE 1}; STOP. A teammate changes the target to 2 and calls the bug fixed. Repair the program and give two tests.

6. Solve the given case

A robot should go from square one to square three. Its program is MOVE ONE, STOP.

Find the cause and repair one thing.

Show your trace, calculation or evidence

Write or draw your reasoning and evidence.

7. Check a changed case

A sorting rule says blue goes left. A blue card goes right. The reader correctly reports blue, but the direction rule says blue goes right. Which part should you fix first?

My independent answer and reason

Write or draw your reasoning and evidence.

Week 12 Test and reflect

8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record · My evidence

Test record: Changed input or condition

My evidence: _____

Test record: Expected answer or fallback

My evidence: _____

Test record: Observed result

My evidence: _____

Test record: Why the result occurred

My evidence: _____

9. Project checkpoint

When the sorter fails, log the first wrong stage and change only the responsible rule, input, or label.

The project change and evidence I will keep

Write or draw your reasoning and evidence.

10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

11. Transfer practice from the original programme

The requirement is to move exactly three squares forward from any valid starting square, then stop.

Starting at square 0, repeat twice {MOVE 1}; STOP ends at 2. A teammate reduces the required distance to two squares. Repair the program without changing the requirement; test starts 0 and 2.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

12. W1 Locate the first difference

A procedure must add 2, double the updated value, then subtract 1. Starting at 3, an observed trace is 3, 5, 8, 7. Write the expected trace, identify the first difference and explain why the final subtraction is not necessarily a separate fault.

My reasoning and evidence

13. W2 Choose between two explanations

In W1 the faulty middle step may add 3, or add the original starting value. Both explain 5 becoming 8 when the start is 3. Choose start 1 or start 3 for a revealing new test. Predict the final result under each explanation and under the intended procedure.

My reasoning and evidence

14. W3 Reject a repair that hides the symptom

A learner repairs the original points machine by reporting 12 for every input, because 12 is correct from start 4. Give a different starting input and its expected output under add 3, double, subtract 2. Explain why the constant report is not a valid repair and what earlier evidence should be preserved.

My reasoning and evidence

15. W4 Debug a new ticket rule

A paper ticket rule must add 1 to the input count. If the updated count is at least 5, report Group; otherwise report Single. A copied version says above 5. Choose the smallest three consecutive nonnegative starting counts that test just below, exactly at and just above the comparison boundary. Predict both versions, identify the revealing start and state the repair and retests.

My reasoning and evidence

16. W5 Make a defensible completion claim

After repairing W4, all three chosen tests pass. Write a short report with the fault, the exact repair, the three inputs and expected outputs, and what those passing tests establish. Explain why saying every possible error is now impossible would go beyond the evidence.

My reasoning and evidence

17. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

18. Pictorial case practice: Debug the Cause

A fictional paper calculator should double the current value, but a copied instruction behaves strangely.

After input 6, the faulty step reports 12. Hypothesis A: add 6. Hypothesis B: double.

Both hypotheses fit this first observation. A new input may be chosen.

A revealing test separates explanations

	Input	Add 6 predicts	Double predicts
Original input	6	12	12
Recorded new input	2	8	4
Proposed input	0	6	0

Read a cell using both its row and column labels.

Debug the Cause

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Observe

Input 6 produced 12; that fact fits two explanations.



2. Predict alternatives

For input 2, adding 6 predicts 8; doubling predicts 4.



3. Run revealing test

The fictional new record is input 2 -> output 8.



4. Locate cause

This record supports add-6 over doubling for this comparison.

CHANGE ONE THING

Use input 6 again instead of 2.

Explain the changed result using the matching lesson.

Propose input 0 and give both predicted outputs without inventing an observed result.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: When the sorter fails, write competing explanations and predicted traces before changing code or instructions.

19. Project portfolio milestone

Project: Helpful Classroom Sorter

Current milestone: Make a transparent first version. Keep a trace of how one input becomes an output; explain the rule or learned component.

When the sorter fails, write competing explanations and predicted traces before changing code or instructions.

Evidence to keep · My record

Evidence to keep: This week’s input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week’s record rather than starting again.

20. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional paper calculator should double the current value, but a copied instruction behaves strangely.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? This record supports add-6 over doubling for this comparison.

1. Observe
2. Run revealing test
3. Predict alternatives
4. Locate cause

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Locate cause This record supports add-6 over doubling for this comparison. Repeating input 6 would not separate the two hypotheses because both again predict 12. Input 2 makes their predictions different before the test is run. The second record supports investigating an added constant, but inspecting the instruction is still needed to confirm the implementation.

Practice 2: explain the deciding evidence

Changed-case practice: Use input 6 again instead of 2. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed input or the procedure.
2. Both hypotheses predict 12; the test remains inconclusive.
3. This one example guarantees correct results for every future input.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: Both hypotheses predict 12; the test remains inconclusive. A revealing test requires different predicted outputs, not merely another recorded run.

Repair a missing building block

Review Repeating Efficiently (ai-1/11). Goal: Trace a loop by testing its condition before every repetition. Short practice: Trace a fresh start of 9. How many additions occur, and must the report equal 10?. Return to this week and explain what you changed.

Review Algorithms as Recipes (ai-1/9). Goal: Trace an ordered algorithm using the current value after every instruction. Short practice: Use the arrows to explain where the two zero-start runs first differ.. Return to this week and explain what you changed.

Review Sound Becomes a Signal (ai-1/6). Goal: Interpret an ordered sequence of sound-signal samples with a stated time interval. Short practice: Does the all-zero retained record prove the full signal was silent? Use the picture.. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
